

WEB PROGRAMMING (CoSc3091) INDIVIDUAL ASSIGNMENT REPORT

Project Title: Student Result Management System

Department: Computer Science

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ID Number: 053/16

Course Code: CoSc3091

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1. Executive Summary & Project Overview

The **Student Result Management System** is a full-stack web application designed and implemented as an individual project for the Web Programming course. The system automates the process of managing academic records, scores, and grades for educational institutions.

It allows administrators to perform administrative duties—specifically managing student records and scores through Create, Read, Update, and Delete (CRUD) operations—while giving students secure, read-only access to view their personalized academic progress.

Key Objectives Met:

- Built a multi-page web application adhering to semantic HTML5 standards.
- Developed a responsive and flexible user interface using mobile-first CSS3 layout techniques (Flexbox and Media Queries).
- Integrated JavaScript for client-side form validation and dynamic layout behavior.
- Developed secure server-side logic in PHP with PDO database connectivity and SQL injection prevention.
- Implemented secure authentication with password hashing (`password_hash()`) and session handling.

2. System Architecture & Tech Stack

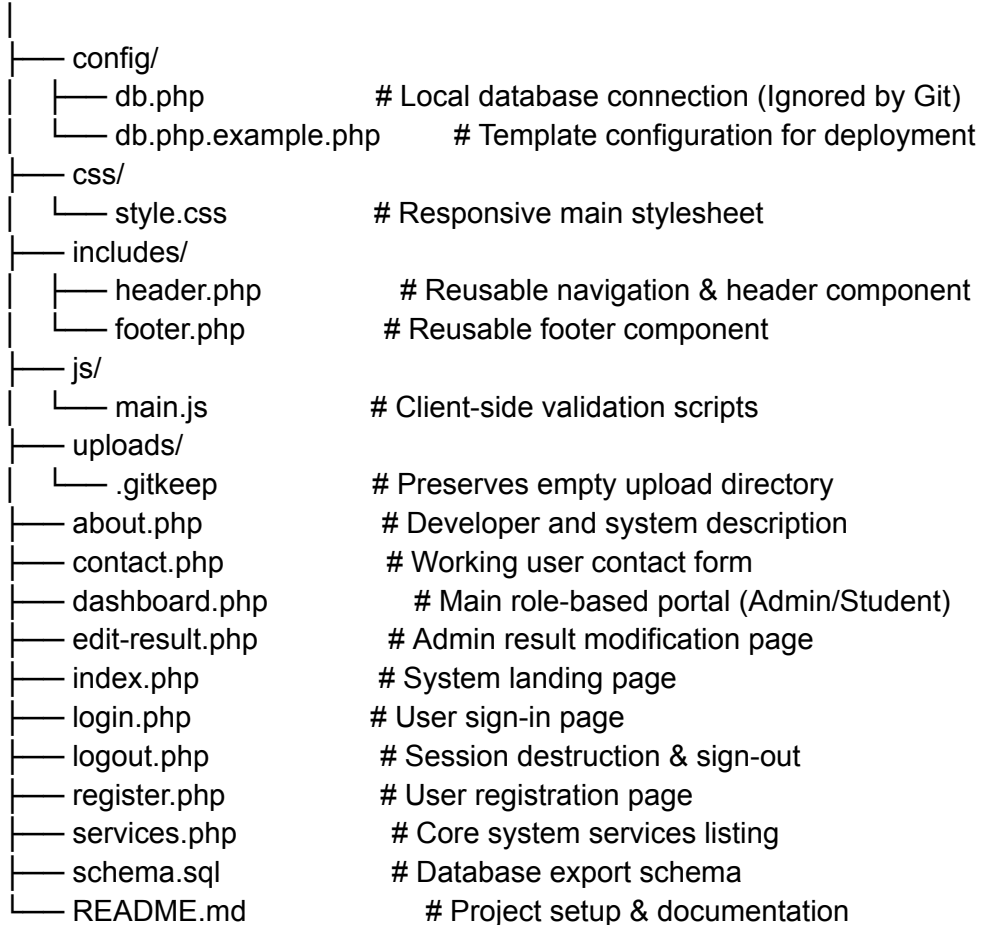
Technology Stack

- **Frontend:** HTML5, CSS3, JavaScript (ES6+).
- **Backend:** PHP 8.x (Modular architecture).
- **Database:** MySQL / MariaDB (Relational Database).
- **Database Access:** PHP Data Objects (PDO) with prepared statements.
- **Version Control:** Git & GitHub.

Directory Structure

The repository follows a clean, modular directory structure ensuring maintainability and separation of concerns:

student-result-system/



3. Database Design & Schema Structure

The application relies on a MySQL relational database named `student_db`. The schema consists of three core tables:

❖ **users Table**

Stores authentication and role information for system users.

- `id` (INT, Primary Key, Auto Increment)
- `full_name` (VARCHAR(100), NOT NULL)
- `email` (VARCHAR(100), UNIQUE, NOT NULL)
- `password` (VARCHAR(255), Hashed using `PASSWORD_BCRYPT`)
- `role` (ENUM('student', 'admin'), DEFAULT 'student')[cite: 1]
- `created_at` (TIMESTAMP)

❖ **results Table**

Stores individual student academic scores[cite: 1].

- `id` (INT, Primary Key, Auto Increment)
- `student_id` (INT, Foreign Key referencing `users(id)` ON DELETE CASCADE)
- `course_code` (VARCHAR(20), NOT NULL)
- `course_name` (VARCHAR(100), NOT NULL)
- `marks` (INT, NOT NULL)
- `grade` (VARCHAR(5), NOT NULL)

❖ **contact_messages Table**

Stores messages submitted through the contact page[cite: 1].

- `id` (INT, Primary Key, Auto Increment)
- `name` (VARCHAR(100), NOT NULL)
- `email` (VARCHAR(100), NOT NULL)
- `message` (TEXT, NOT NULL)
- `created_at` (TIMESTAMP)

4. Key Functional Features & Implementation

❖ **User Authentication & Authorization:**

- User registration encrypts passwords with `password_hash()` before database insertion.
- Login validates credentials using `password_verify()` and establishes PHP `$_SESSION` state.
- Access control enforces role restrictions (Students cannot access result editing endpoints).

❖ **Result CRUD Functionality:**

- **Create:** Admin adds new subject scores for students.
- **Read:** Dynamic table rendering based on active session role.

- **Update:** Admin modifies existing marks via [edit-result.php](#).
- **Delete:** Admin removes obsolete records with JS confirmation popups.
- ❖ **Responsive UI Layout:**
 - Designed with CSS media queries ensuring tables overflow gracefully with horizontal scrolling on smaller mobile viewports.
 - Compact action button styling for clean mobile presentation.

5. System Screenshots

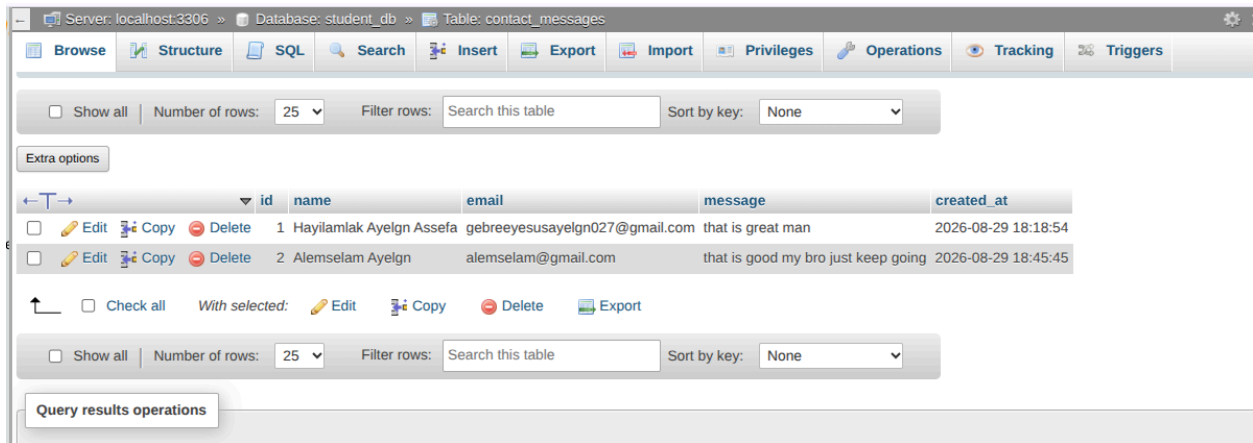
❖ users

	id	full_name	email	password	role	created_at
☐ Edit Copy Delete	1	Hayilamlak Ayelgn Assefa	gebreeyesusayelgn027@gmail.com	\$2y\$10\$ofMc7a23cTFHtrk4YNtPne3gameB.rxkJlBMyLpYT...	student	2026-08-29 18:19:54
☐ Edit Copy Delete	2	Mulubrhan Ayelgn	mulubrhan@gmail.com	\$2y\$10\$II.PK3n5hJaGjLvYN1ByX.p1hj3qZQ6qzI8Ej0ywyXy...	student	2026-08-29 18:34:52
☐ Edit Copy Delete	3	Alemselam	alemselam@gmail.com	\$2y\$10\$OrePp2BxZfNd6p9Ui4pINO2YQVPunjwxGhSCa6BwmMv...	student	2026-08-29 18:46:07
☐ Edit Copy Delete	4	Hayilamlak	admin@gmail.com	\$2y\$10\$qVqYXis.Q.3R2OVvbKW.j.DZk.yVKnRxegcpBCR1udM...	admin	2026-08-30 15:08:16
☐ Edit Copy Delete	5	Alex	student@example.com	\$2y\$10\$7hwIzUYuHbWSKuVmTRPx.Ff57rpR.PZ4NEnabBliLLG...	student	2026-08-30 15:14:00

❖ Result

	id	student_id	course_code	course_name	marks	grade
☐ Edit Copy Delete	1	2	046/16	Javascript	95	A+
☐ Edit Copy Delete	2	3	022/16	PHP	85	A
☐ Edit Copy Delete	3	1	053/16	JavaScript & PHP	72	A+
☐ Edit Copy Delete	4	5	012/16	Java	85	A

❖ *contact_messages*



Server: localhost:3306 » Database: student_db » Table: contact_messages

Extra options

	id	name	email	message	created_at
☐ Edit Copy Delete	1	Hayilamlak Ayelgn Assefa	gebreeyesusayelgn027@gmail.com	that is great man	2026-08-29 18:18:54
☐ Edit Copy Delete	2	Alemselem Ayelgn	alemselem@gmail.com	that is good my bro just keep going	2026-08-29 18:45:45

Query results operations

1. **Figure 1:** Home Landing Page ([index.php](#))
2. **Figure 2:** Login & Authentication Interface ([login.php](#))
3. **Figure 3:** Admin Dashboard with Result Management ([dashboard.php](#))
4. **Figure 4:** Mobile Viewport Responsive Layout

6. Individual Contribution & Originality Statement

I, **Haylamlak Gebreeyesus**, declare that this project, titled **Student Result Management System**, is my own original work developed for the Web Programming course. All frontend styles, backend PHP logic, database structures, and Git history reflect my individual effort in fulfilling the assignment specifications.

GitHub Repository: <https://github.com/Haylamlak/student-result-system>

Test Admin Account: [admin@example.com](#) | Password: [admin123](#)

Test Student Account: [student@example.com](#) | Password: [student123](#)